



BSER-Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models

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Abstract. Speech-emotion recognition,(SER) is an important category in human-computer interaction as it makes it possible for a machine to comprehend and interpret human emotions. It seeks to determine the categories of emotions elicited through speech signals during social communication. This study focuses on the impact of feature extraction techniques and machine learning models on SER performance, utilizing a combined dataset of the SUBESCO and BanglaSER datasets and individual also. Data augmentation through noise injection enhances model robustness. Mel-Frequency Cepstral Coefficients (MFCCs) are extracted as primary features, with a sensitivity analysis conducted by varying the number of MFCCs (1 to 50) to determine the optimal value for SER accuracy. The impact of different sampling rates on feature extraction and subsequent model performance is also investigated. Various machine learning models, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest, and Logistic Regression, are trained and evaluated. The sensitivity analysis reveals that using 30 MFCCs yields the highest accuracy, and higher-order MFCCs significantly improve model performance in BSER. The optimal sampling rate for the dataset is found to be 44,100 Hz. Among the evaluated models, KNN demonstrates the best performance with an accuracy of 92.10%. This study provides important insights regarding the practical application of machine learning for SER in the context of the Bengali language.

Keywords: SER, MFCC, Data augmentation, KNN, Machine learning, sampling rate

1. Introduction

Identifying human emotions from voice signals through machine learning techniques is crucial for developing more intuitive and natural human-computer interaction (HCI) systems. The integration of signal processing and artificial intelligence has become a significant trend in SER [1], with research on effective systems underway for over two decades [2]. Research indicates a connection between human vocal tone and emotion, with computer scientists emphasizing the importance of the SER system (B. W. Schul. SER has broad applications that offer considerable advantages across various fields, including HCI [3], mental stress [4], schooling [5], virtual assistant, and robotics [6]. Despite ongoing research into numerous techniques, challenges remain in computational efficiency, generalization, and real-world applicability. Traditional ML methods, such as SVM and RF, have been used for SER [7], but they exhibit limitations in generalizing unseen data. In this study, we explored the SER system with a developed K-NN model. However, selecting the appropriate features is a critical element of the design process. Research indicates that commonly used acoustic features include MFCC, zero-crossing rate (ZCR), pitch energy, log-Mel spectrogram, root mean square (RMS), log frequency power coefficients, and linear predictive coding. This study addresses existing gaps by introducing a pipeline specifically designed for the Bengali language. Our study utilizes MFCC features and the K-NN model. To overcome challenges related to limited datasets, the proposed methodology implements merged datasets of the SUBSESCO and BalngaSER datasets. We also employed the data augmentation technique by adding noise to expand the dataset and strengthen the model. This study also analyzes how the accuracy changes with sampling rate variation, and a comparative analysis between K-NN, SVM, RF, & LR was done.

2. Literature Review

Speech Emotion Recognition (SER) has garnered significant attention in recent years due to its wide range of applications in human-computer interaction, mental health monitoring, and virtual assistants. While extensive research has been conducted in widely spoken languages such as English, German, and Mandarin, SER research in Bengali remains relatively underexplored. This section reviews previous studies in SER across various languages, with a particular emphasis on Bengali SER, highlighting the contributions and limitations of existing works and positioning the current study within this broader context. SER systems have been extensively developed for high-resource languages like English and German. Traditional machine learning methods, such as SVM and RF, have been widely used with acoustic features like MFCC, ZCR, and pitch energy. Deep learning approaches, including CNN and Long Short-Term Memory (LSTM) networks, have also shown promising results by capturing temporal and spectral dependencies in speech signals [1]. However, these methods often require large datasets and computational resources, which limit their applicability to low-resource languages. Despite Bengali being the seventh most spoken language globally, SER research in Bengali is still in its nascent stages. Early studies in Bengali SER relied on small datasets and simple classifiers, such as SVM and KNN [8]. The introduction of high-quality datasets like SUBESCO [9] and BanglaSER [10] has significantly

facilitated research in this area. Recent works have explored both traditional and deep learning methods for Bengali SER. For instance, Ahmed et al. (2023) [11] proposed an ensemble 1D-CNN-LSTM-GRU model with data augmentation, achieving notable improvements in accuracy. However, deep learning models often struggle with overfitting and require large amounts of training data, which are not always available for Bengali. Data augmentation has emerged as a vital technique to address the scarcity of annotated Bengali speech data. Noise injection and pitch-shifting methods have been employed to expand the training set and enhance model generalization [12]. However, the effectiveness of these methods in Bengali SER remains underexplored. Unlike previous works that focused on individual datasets, this study combines two prominent Bengali emotional speech datasets—SUBESCO and BanglaSER—to create a more diverse and comprehensive dataset. The use of MFCC features, along with data augmentation through noise injection, further enhances the model's ability to generalize to unseen data.

3 Methodology

3.1 Dataset

The SUBESCO [9] and the BanglaSER [10] datasets have been used for our research project. The SUBESCO dataset comprises 7,000 audio files, equally distributed across seven emotions: fear, surprise, anger, neutral, sadness, disgust, and happiness. The BanglaSER dataset, on the other hand, consists of 1,467 audio files covering five emotions. Both datasets include recordings from male and female speakers to ensure gender balance, but there are inherent demographic biases, as most participants are native Bengali speakers from specific regions of Bangladesh. We merged both of these datasets and conducted our research.

3.2 Data Augmentation and Feature Extraction

Data augmentation [13] is used to increase the dataset and prepare this for the real world, we added noise [14] to expand the dataset. We used MFCC [15] in the feature extraction process. These features effectively captured the spectral characteristics of speech, enabling the model to distinguish between different emotions. MFCC helped in representing human auditory perception.

3.3 Data preprocessing

In the audio dataset, we assigned 0 to the null spaces. Standard Scaler [16] is used for scaling data equally. This normalization ensures that all features are equally contributing to the model, preventing any dominant influence from higher-magnitude features.

3.4 Machine learning models

KNN, SVC, LR, and RF from `cuml` [17] are used for building the model. KNN classifies based on the majority votes among the nearest neighbors, while SVM finds an optimal hyperplane for the separation. Proper hyperparameters for the SVM model are found using grid search in model building to maintain the overall balance of the model. LR predicts using the probability of class membership, and RF is an ensemble of decision trees that improves accuracy and reduces over-fitting problems.

4. Results and Discussions

Our research shows the effectiveness of KNN for BSER using MFCCs as the main feature. This study employs a combined dataset from SUBESCO and BanglaSER. We analyzed the effects of varying MFCC values and different sampling rates while comparing KNN with other machine-learning models, including SVM, LR, and RF. Figure 2 illustrates the relationship between the number of MFCC features and the accuracy achieved by our model in both the testing and training phases. As the number of MFCCs increases from 1 to 50, the accuracy consistently rises, reaching a peak of 92.10% with 30 MFCCs. These results indicate that higher-order MFCCs significantly enhance the model's ability, improving its efficacy in distinguishing various emotions, but excessive extraction may introduce redundancy.

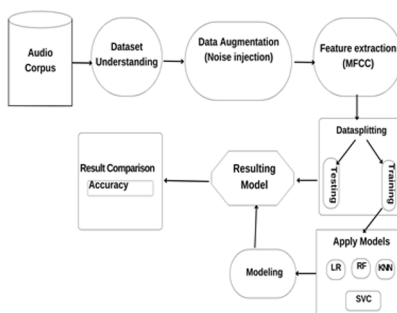


Fig.1. Flow chart of our works

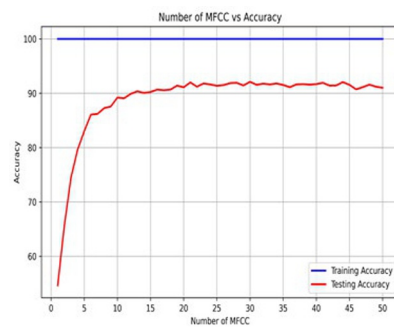


Fig.2. Number of MFCC vs Accuracy

Table 1: Variations of Accuracy to sampling rate

Number of MFCC	Sampling Rate (Hz)	Model	Training Accuracy	Testing Accuracy
50	16,000	KNN	100	88.83
50	22,050	KNN	100	89.42
50	32,000	KNN	100	90.75
50	44,000	KNN	100	90.87
50	44,100	KNN	100	91.00

Table 2: Comparison between ML models

Number of MFCC	Sampling Rate (Hz)	Model	Training Accuracy	Testing Accuracy
50	44,100	KNN	100	91.00
50	44,100	SVC	100	63.40
50	44,100	LR	100	73.52
50	44,000	RF	100	78.62

Table 1 illustrates how accuracy varies with different sampling rates for the KNN algorithm using 50 MFCCs as the standard value. The optimal performance was achieved at a sampling rate of 44,100 Hz, resulting in a testing accuracy of 91.00. Higher sampling rate marginally increases the testing accuracy to 91.28%, with a significant rise in computational costs. Table 2 presents a comparative overview of the performance of the KNN algorithm in comparison to the SVC, LR, and RF under standardized conditions, including the use of 50 MFCCs, a sampling rate of 44,100 Hz, and data augmentation techniques. KNN achieved the highest testing accuracy of 91.00%. We also analyzed the performance of the KNN algorithm for the SUBESCO and BanglaSER datasets separately, while having the same experimental setup. Our model achieved an accuracy of 90.50% with the SUBESCO dataset and 87.96% with BanglaSER.

5. Conclusion and Future Works

This study demonstrates the effectiveness of the KNN algorithm as a reliable and high-performing model for BSER. The results indicate that KNN excels in capturing the non-linear relationships present in emotional speech data, making it the most suitable choice among the models evaluated. Our analysis of global Speech Emotion Recognition (SER) studies indicates that Bengali SER has unique challenges and opportunities. Unlike the dominant deep learning methods in English and German, traditional models like K-Nearest Neighbors (KNN) remain effective for Bengali speech due to its specific linguistic features requiring non-linear classifiers. The analysis also underscores the benefits of combining datasets like SUBESCO and BanglaSER. While the findings are promising, there is significant potential for future improvements. Additionally, incorporating multi-modal data could help create a more comprehensive emotion

recognition system. Refining noise-reduction techniques and ensuring robustness in noisy environments are critical areas for further research.

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References

1. R. Jahangir, Y. W. Teh, F. Hanif, and G. Mujtaba, "Deep learning approaches for speech emotion recognition: State of the art and research challenges," *Multimedia Tools and Applications*, vol. 80, no. 16, pp. 23 745–23 812, (2021).
2. B. W. Schuller, "Speech emotion recognition: Two decades in a nutshell, benchmarks, and ongoing trends," *Communications of the ACM*, vol. 61, no. 5, pp. 90–99, (2018).
3. M. U. Islam and B. M. Chaudhry, "A framework to enhance user experience of older adults with speech-based intelligent personal assistants," *IEEE Access*, vol. 11, pp. 16 683–16 699, (2022).
4. M. Lech and L. He, "Stress and emotion recognition using acoustic speech analysis," in *Mental Health Informatics*. Springer, 2013, pp. 163–184.
5. Q. Luo and H. Tan, "Facial and speech recognition emotion in distance education system," in *The 2007 International Conference on Intelligent Pervasive Computing (IPC 2007)*. IEEE, 2007, pp. 483–486, (2007).
6. J. G. Razuri, D. Sundgren, R. Rahmani, A. Moran, I. Bonet, and ´ A. Larsson, "Speech emotion recognition in emotional feedback for human-robot interaction," *International Journal of Advanced Research in Artificial Intelligence (IJARAI)*, vol. 4, no. 2, pp. 20–27, (2015).
7. L. Zheng, Q. Li, H. Ban, and S. Liu, "Speech emotion recognition based on convolution neural network combined with random forest," in *2018 Chinese control and decision conference (CCDC)*. IEEE, 2018, pp. 4143–4147, (2018).
8. Sinha, M., Dasgupta, T., & Basu, A. (2014, August). Influence of target reader background and text features on text readability in bangla: A computational approach. In *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers* (pp. 345-354), (2014).
9. S. Sultana, M. S. Rahman, M. R. Selim, and M. Z. Iqbal, "Sust bangla emotional speech corpus (subesco): An audio-only emotional speech corpus for bangla," *Plos one*, vol. 16, no. 4, p. e0250173, (2021).
10. R. Das, N. Islam, M. Ahmed, S. Islam, S. Shatabda, and A. Islam, "Banglaser: A speech emotion recognition dataset for the bangla language," *Data in Brief*, vol. 42, p. 108091, 03 (2022).

11. Ahmed, M. R., Islam, S., Islam, A. M., & Shatabda, S. An ensemble 1D-CNN-LSTM-GRU model with data augmentation for speech emotion recognition. *Expert Systems with Applications*, 218, 119633, (2023)..
12. Banshal, S. K., Das, S., Shammi, S. A., & Chakraborty, N. R.. MONOVAB: An Annotated Corpus for Bangla Multi-label Emotion Detection. *arXiv preprint arXiv:2309.15670*, (2023).
13. M. Xu, F. Zhang, X. Cui, and W. Zhang, "Speech emotion recognition with multiscale area attention and data augmentation," in ICASSP 2021- 2021 IEEE International conference on acoustics, speech and signal processing (ICASSP). IEEE, 2021, pp. 6319–6323, (2021).
14. M. R. Ahmed, S. Islam, A. M. Islam, and S. Shatabda, "An ensemble 1d-cnn-lstm-gru model with data augmentation for speech emotion recognition," *Expert Systems with Applications*, vol. 218, p. 119633, (2023).
15. C. Ittichaichareon, S. Suksri, and T. Yingthawornsuk, "Speech recognition using mfcc," in International conference on computer graphics, simulation and modeling, vol. 9, (2012).
16. Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort et al. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* 12, null (2/1/2011), 2825–2830, (2011).
17. Raschka, S., Patterson, J., & Nolet, C. Machine learning in python: Main developments and technology trends in data science, machine learning, and artificial intelligence. *Information*, 11(4), 193, (2020)..

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